DISCRETE MATH · ELEMENTARY NUMBER THEORY AND METHODS OF PROOF

Euclidean Algorithm Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Euclidean Algorithm Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. Without the tool, guess gcd(84, 36). How did you guess?
   12 (for example by listing divisors or factoring $84 = 2^2 \cdot 3 \cdot 7$ and $36 = 2^2 \cdot 3^2$).
2. If $a = bq + r$, why should a and b have the same common divisors as b and r?
   Anything dividing a and b divides $r = a - bq$; anything dividing b and r divides $a = bq + r$.

2 Explore use the tool

1. a = 252, b = 198. Step through the cuts. Record each equation $a = bq + r$.
   $252 = 198 \cdot 1 + 54$; $198 = 54 \cdot 3 + 36$; $54 = 36 \cdot 1 + 18$; $36 = 18 \cdot 2 + 0$. gcd = 18.
2. What is the side of the smallest squares? How does it relate to the last equation?
   18, the last nonzero remainder (the divisor in the step whose remainder is 0).
3. Find a pair under 100 that needs at least 6 steps. What kind of numbers are they?
   For example 89 and 55 (9 steps) or 55 and 34 (8 steps): consecutive Fibonacci numbers.
4. Use the Quotient and remainder tab with n = −23, d = 7. Record q and r.
   q = −4, r = 5, since $-23 = 7(-4) + 5$.

3 Record use the tool and fill in the table

GCD, LCM and a linear combination

Use Show all steps. Check that gcd · lcm = ab.

a	b	Steps	gcd(a, b)	lcm(a, b)	gcd = s·a + t·b
84	36	2	12	252	$12 = 1 \cdot 84 + (-2) \cdot 36$
91	120	4	1	10920	$1 = (-29) \cdot 91 + 22 \cdot 120$
72	60	2	12	360	$12 = 1 \cdot 72 + (-1) \cdot 60$
89	55	9	1	4895	$1 = (-21) \cdot 89 + 34 \cdot 55$

4 Explain what did you discover?

1. gcd(a, b) = gcd(b, r), where r is the remainder when a is divided by b.
2. The GCD is the last nonzero remainder.
3. gcd(a, b) · lcm(a, b) = ab.
4. For any integer n and $d > 0$, $n = dq + r$ with 0 ≤ r < d.

5. Why must the Euclidean algorithm stop? Use the remainders in your answer.
   Each remainder is a nonnegative integer smaller than the one before (r < b), so the remainders strictly decrease and must reach 0 after finitely many steps.

5 Apply a new situation, no tool

1. Use the Euclidean algorithm to find gcd(1071, 462).
   1071 = 462·2 + 147; 462 = 147·3 + 21; 147 = 21·7 + 0. gcd = 21.
2. Find lcm(1071, 462).
   1071 · 462 / 21 = 23562.
3. Write gcd(30, 21) as a combination 30s + 21t.
   30 = 21·1 + 9, 21 = 9·2 + 3, so 3 = 21 − 2·9 = 21 − 2(30 − 21) = 3·21 − 2·30: s = −2, t = 3.
4. Find −40 div 6 and −40 mod 6.
   −40 = 6(−7) + 2, so −40 div 6 = −7 and −40 mod 6 = 2.

Teacher notes

Intended discovery: Replacing (a, b) by (b, r) keeps the common divisors, so the last nonzero remainder is the GCD; back-substitution gives Bézout coefficients; gcd·lcm = ab; remainders are always nonnegative.

Likely misconceptions:

- Taking the last remainder (0) instead of the last nonzero remainder.
- Writing −23 = 7(−3) − 2 (negative remainder).
- Thinking large numbers always need many steps (Fibonacci pairs are slowest).

Answers that can vary: Linear-combination coefficients are not unique (adding b/g to s and subtracting a/g from t gives another pair); the tool shows one pair from the table method. Explore 3 pairs vary.